

PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)
(Estd. under the Panjab University Act VII of 1947 -enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc (HARDWARE AND NETWORKING)
as per NEP 2020

w.e.f session 2026-27

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BHN-101	Core Discipline	Basics of Computer System	4	--	4	80	20	--	--	100
BHN-102	Core Discipline	Fundamental of PC Hardware	2	--	4	40	10	--	--	100
BHN-102P		Practical Based on BHN-102	--	2		--	--	40	10	
BHN-103	Core Discipline	PC Configuration	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I & Viva- Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BHN-201	Core Discipline	Data Communication and Networks	4	--	4	80	20	--	--	100
BHN-202	Core Discipline	Computing Software	2	--	4	40	10	--	--	100
BHN-202P		Practical Based on BHN-202	--	2		--	--	40	10	
BHN-203	Core Discipline	Network Configuration	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	--	4	4	--	--	80	20	100
Total Credits					**24**					**600**

Job Role - Network Support Engineer

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 5th Semester. Viva-Voce examination to be held within 3-weeks of the start of 5th semester.

** Minimum Credits Required to get Diploma Course in (HARDWARE AND NETWORKING) =48 + 4 #.
#Exit 2: Diploma Course-Student can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the second year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

***One credit of theory lecture is of 1 hour duration and one credit of practical is of 2 hours duration.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BHN-301	Core Discipline	Basics of Routing and Switching	4	--	4	80	20	--	--	100
BHN-302	Core Discipline	Programming using C Language	2	--	4	40	10	--	--	100
BHN-302P		Practical Based on BHN-302	--	2		--	--	40	10	
BHN-303	Core Discipline	Routing Configuration	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 301	Multidisciplinary Course	Entrepreneurship Development[3]	3	-	3	60	15	-	-	75
SEC III	Skill Enhancement Course	SEC Options[2]			3					75
VAC II	Value-Added Course	VAC Options[3]			2					50
PW II	Project Work	Project Work-II & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BHN-401	Core Discipline	Digital Electronics and Computer Organisation	4	--	4	80	20	--	--	100
BHN-402	Core Discipline	Network Administration	2	--	4	40	10	--	--	100
BHN-402P		Practical Based on BHN-402	--	2		--	--	40	10	
BHN-403	Core Discipline	DOS and Windows Operating System	2	--	4	40	10	--	--	100
BHN-403P		Practical Based on BHN-403	--	2		--	--	40	10	
BHN-404	Core Discipline	Switching Configuration	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
VAC III	Value-Added Course	VAC Options[3]			2					50
SIT 401	Internship	Summer Industrial Training (SIT)*	--	4	4	--	--	80	20	100
Total Credits					**24**					**600**

Job Role - Network Support Engineer

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 5th Semester. Viva-Voce examination to be held within 3-weeks of the start of 5th semester.

** Minimum Credits Required to get Diploma Course in (HARDWARE AND NETWORKING) =96 + 4 #.
#Exit 2: Diploma Course-Student can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the second year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

***One credit of theory lecture is of 1 hour duration and one credit of practical is of 2 hours duration.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

Semester-V

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BHN-501	Core Discipline	Information System Design	4	--	4	80	20	--	--	100
BHN-502	Core Discipline	Wireless Communication and Networking	4	--	4	80	20	--	--	100
BHN-503	Core Discipline	Linux Operating System	2	--	4	40	10	--	--	100
BHN-503P		Practical Based on BHN-503	--	2		--	--	40	10	
BHN-504	Core Discipline	Web Designing	--	4	4	--	--	80	20	100
BHN-505	Core Discipline	Advance Routing and Switching	--	4	4	--	--	80	20	100
VAC IV	Value-Added Course	Common Basket			2					50
PW III	Project Work	Project Work-III & Viva-Voce			2					50
Total Credits					**24**					**600**

Semester-VI

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BHN-601	Core Discipline	Internet of Things (IoT) in Networking	4	--	4	80	20	--	--	100
BHN-602	Core Discipline	Cyber Security and Data Protection	4	--	4	80	20	--	--	100
BHN-603	Core Discipline	Firewall Technology	2	--	4	40	10	--	--	100
BHN-603P		Practical Based on BHN-603	--	2		--	--	40	10	
BHN-604	Core Discipline	Artificial Intelligence and Data Science	2	--	4	40	10	--	--	100
BHN-604P		Practical Based on BHN-604	--	2		--	--	40	10	
BHN-605	Core Discipline	Core Python Concepts	--	4	4	--	--	80	20	100
BHN-606	Core Discipline	Advance IP Routing	--	4	4	--	--	80	20	100
Total Credits					**24**					**600**
Job Role - Network Administrator										

***One credit of theory lecture is of 1 hour duration and one credit of practical is of 2 hours duration.

1. Skill Enhancement Courses (SEC): Annexure I
2. Multidisciplinary Courses (MDSC): Annexure II
3. Value Added Courses (VAC): Annexure III

ESSENTIAL COMPONENTS OF COURSE OF
B.Voc (HARDWARE AND NETWORKING)
(SEMESTERS I & II)

PROGRAMME OUTCOMES

1. The primary objective of this course is to prepare students for careers as Field Technicians (Computing and Peripherals) and Hardware and Network Engineers/Administrators. These roles are crucial in the IT industry, particularly in servicing firms such as telecom companies and equipment manufacturing firms. Students will gain the necessary skills and knowledge to excel in these positions.

2. The course places a strong emphasis on practical, hands-on training. It recognizes the importance of providing students with an early platform to manage and troubleshoot complex failures in computer hardware and networks. By engaging in practical exercises, students will develop the skills required to analyze problems and implement effective solutions, fostering a deeper understanding of the field.

3. A key focus of the curriculum is to cultivate the ability of students to identify and resolve issues related to computer hardware and networks. The course structure is designed to equip students with problem-solving skills, enabling them to troubleshoot and address challenges in a real-world context. This emphasis on practical problem-solving enhances the employability of graduates.

4. The curriculum is well-rounded, encompassing both general education and skill development components. Beyond technical expertise, students will receive a broad education that includes relevant industry knowledge and soft skills. This comprehensive approach ensures that graduates are not only technically proficient but also well-rounded professionals capable of adapting to diverse work environments.

5. To bridge the gap between theoretical knowledge and practical application, the course integrates extensive practical and on-the-job training components. Students will have the opportunity to apply their theoretical learning in real-world scenarios, preparing them for the challenges they may encounter in their future roles. This hands-on experience is invaluable for building confidence and competence.

6. To further connect students with the professional world, the course incorporates industry interactions. Students will have the chance to engage with professionals in the field, participate in workshops, and attend seminars. This exposure facilitates networking opportunities, allowing students to learn from industry experts, stay updated on industry trends, and potentially secure internships or job placements. This connection to the industry is instrumental in preparing students for a successful transition from education to the workforce.

Semester I

<table>
<tr><th colspan="2">Course Title</th><th>Basics of Computer System</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-101</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 4 marks each.
iv. The students are required to attempt one long question of 16 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Field Technician- Computing & Peripherals</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>The course aims to equip students with a profound understanding of computer fundamentals, hardware, software, and networking. To empowers the ability to analyze advantages and disadvantages, design computer system architectures, apply input and output devices, and comprehend emerging technologies and well-rounded expertise in basics concepts.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Demonstrate a comprehensive understanding of computer fundamentals, including the historical development and characteristics of computers.
CO2. Apply knowledge of input and output devices, computer memory, and different types of software in practical scenarios.
CO3. Analyze and evaluate the functions and types of operating systems, data processing systems, and translators.
CO4. Demonstrate proficiency in understanding computer networks, the Internet, and emerging technologies, with the ability to classify and describe different types of networks.</td></tr>
</table>

9	**Pedagogy of the Course Work**	Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.
10	**Course Content** **UNIT- I** Computers: Characteristics, Historical development of computers, Evolution of computing devices, Generations of Computers: First Generation (Vacuum Tubes, Batch Processing), Second Generation (Transistors), Third Generation (Integrated Circuits, Time-Sharing Systems), Fourth Generation (Microprocessors, Personal computers), Fifth Generation (Artificial intelligence). Hardware, Software and Firmware, Data and Information. Computer Codes: Decimal System, Binary System, Hexadecimal System, Octal System and ASCII Code. Decimal to Binary conversion, Binary to Decimal Conversion. **UNIT- II** Introduction to Computer Networks: Definition and Importance of Computer Networks, Basic Components of a Computer Network, Classification of Networks, Network Topologies. Internet: Historical Overview of The Internet, Evolution of The World Wide Web (WWW), WWW Components, Internet Governance and Standards, Internet Services and Applications (Email Protocols, Telnet, FTP, IRC, Usenet News). Internet Access Methods (Dial-Up, DSL, Cable, Fiber-Optic), Browser Technologies and Features. **UNIT- III** Computer Virus: Types of Viruses, Impact of Virus, Virus Detection and Prevention techniques, Malware. Antivirus and Its Types. Multimedia: Hardware and Software for Multimedia, Multimedia: Component of Multimedia, Basic File Formats, Audio File Formats, Image File Formats, Video File Formats. Multimedia Storage Devices. **UNIT-IV** Emerging Technologies: Basics of Internet of Things (IoT), Components of IoT, Applications of IoT, Basics of Parallel Computing and Distributed Computing, Difference between Parallel computing and Distributed Computing, Advantages and Limitations of Parallel computing and Distributed Computing, Cloud Computing: Component of Cloud Computing, Types of Cloud Computing, and their applications.	
11	**Suggested Resources for Reading** 1. Fundamentals of Computers, V. Rajaraman, PHI Publication. 2. Computer Fundamentals, P. K. Sinha, BPB Publication. 3. Computers Today, Basandra, S. K., Galgotia Publications. 4. Fundamentals of Information Technology, Anshuman Sharma, Lakhanpal Publications. 5. PC Hardware: The Complete Reference, Craig Zacker., TMH Publication. **Note:** Latest edition of text books may be used.	

<table>
<tr><th colspan="2">Course Title</th><th>Fundamental of PC Hardware</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-102</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practicals
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4X 2 marks) = 8 marks
(4X8 marks) = 32 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Field Technician- Computing & Peripherals</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course aims to provide comprehensive knowledge of computer peripherals, hardware components, and system management. Hardware Fundamentals enable to learn about hardware technologies, their implementation/installation, uses, troubleshooting and maintenance.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Demonstrate a comprehensive understanding of computer hardware components, peripherals, and their functionalities.
CO2. Enable the students to learn about hardware technologies and their implementation/installation, uses, troubleshooting and maintenance.
CO3. Install, update, and troubleshoot drivers for various hardware components, including network adapters, sound cards, and graphics cards.
CO4. Configure system settings, understand the role of BIOS, and troubleshoot common BIOS-related issues.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.</td></tr>
</table>

10	**Course Content** **UNIT-I** Peripheral Devices: Introduction to Peripheral Devices, Functionality, Use and Types of Various Peripheral Devices (Mouse, Keyboard, Printer, and Monitor). Add-On Cards and Its Functionality: Network Interface Card, Sound Card and Graphics Card. Uninterruptible Power Supply (UPS): Features, Types (Offline & Online UPS), Advantages & Limitations of UPS. Mother Board: Definition, Importance, Basic Components, Layout of Motherboard, Types of Motherboards (AT, ATX, and BTX). **UNIT-II** Ports: Definition, Role of Ports in Computer Systems, Functions, Types of Ports: Serial Ports and its Types, Parallel Ports, and its Types. Hard Disk: Definition, Importance of Hard Disk, Structure of Hard Disk (Head, Platter, Spindle, Actuator, Tracks, Sectors), Types of Hard Disk (PATA, SATA, SCSI). RAM: Definition, Uses, Types of RAM (SRAM, DRAM, SDRAM). Power Supply Unit: Functions, Components, Power Connectors, Power Requirements for various Computer Components. Processor: Definition, Architecture, Functions, Clock Speed, Cores, and Cache Memory. **UNIT-III** Cables: Importance of Cables in Computer Systems, Types (VGA, DVI, RJ11, RJ45, HDMI, Serial, Parallel Cables, USB (Versions, Specifications, Common Uses, Connectors and Compatibility). Computer Cable Management: Importance of Cable Management for Airflow and Aesthetics, Techniques for Organizing and Securing Cables, Computer Case Management: Types, Form Factors, Importance of Airflow and Ventilation in Computer Cases, Reducing Cable Clutter Inside the Computer Case. Drivers: Definition, Role of Device Drivers, Importance of Drivers for Hardware Functionality, Find and Install Device Drivers. Types of Drivers (Device Drivers, LAN Drivers, Sound Drivers, Graphics Drivers). Cabinet: Role of Cabinet, Types of Computer Cabinets. **UNIT-IV** BIOS: Definition, Importance, Functions, Responsibilities, Role of BIOS in the Boot Process, Importance of BIOS Updates, Procedure for Updating BIOS, Risks and Precautions during BIOS Updates, Common BIOS Error Messages. Antivirus Software: Definition, Features, Significance, Role of Antivirus in Security, Installing and Configuring any Third-Party Antivirus for Optimal Protection. Firewall: Definition, Purpose of Firewalls, Importance in Network Security, Hardware Vs. Software Firewalls, Types of Firewalls (Packet-Filtering Firewalls, Stateful Inspection Firewalls, Proxy Firewalls and Application-Layer Filtering), Limitations and Challenges in Firewall Protection, Real World Examples of Successful Firewall Applications.
11	**Suggested Resources for Reading** 1. The Indispensable PC Hardware Book (4th Edition), Hans-Peter Messmer, Addison Wesley. 2. PC Hardware: A Beginner's Guide by Ron Gilster, McGraw-Hill Education. 3. Computer Hardware and Practical on Networking, Osheen Sharma and Shruti, Mohindra Publishing House. 4. PC Hardware: The Complete Reference, Craig Zacker., TMH Publication. 5. How Computers Work, Ron White and Timothy Edward Downs, QUE. 6. Fundamentals of Computers, V. Rajaraman, PHI Publication. **Note**: Latest edition of text books may be used.
BHN-102P (Practical Based on BHN-102) Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation. 1. Examiner will set total of four questions covering the whole syllabus. 2. Student will attempt any three questions. 3. Each question carries 5 marks. 4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.	

<table>
<tr><th colspan="2">Course Title</th><th>PC Configuration</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-103</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>120 hours of practicals
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>-</td><td>-</td><td>80</td><td>20</td></tr>
</table>
i. The exam of this course will be of 3hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
Instructions for paper setters and candidates:
i. Examiner will set total of four questions covering the whole syllabus.
ii. Student will attempt any three questions.
iii. Each question carries 10 marks.
iv. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Field Technician- Computing & Peripherals</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>The course aims to provide a comprehensive understanding of computer systems, covering fundamental hardware components. It develops practical skills in connecting peripherals, analyze and solve hardware issues logically, and assemble, configure, troubleshoot, and maintain functional computer systems. Additionally, the course focuses on installing Windows, managing drivers, and implementing antivirus software for system security.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Define and explain the fundamental components of a computer system, demonstrating a holistic understanding of hardware architecture.
CO2. Demonstrate practical skills in connecting and configuring peripherals, including mouse, keyboard, display devices, scanning devices, and hard disk.
CO3. Develop critical thinking skills to analyze and solve hardware-related problems, applying a logical and systematic approach to diagnose and address issues.
CO4. Assemble, configure, troubleshoot, and maintain a functional computer system, integrating knowledge from all units for practical application.
CO5. Successfully install the Windows operating system, manage the installation and updating of drivers, install and update anti-virus software.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td></td><td colspan="2">Course Content
UNIT-I
Introduction To Computer Hardware:
• Overview of Computer Components
• Basic Understanding of CPU, RAM, Storage Devices, Various Ports, Slots (Add On Devices) and Input/Output Devices (Keyboard, Mouse, Monitor, Printer).
• Hands-On Practice Connecting and Configuring Peripherals (Mouse, Keyboard, Display Devices, Scanning Devices, Hard Disk).</td></tr>
</table>

	• Studying Various Components of The Motherboard. • Introduction to Hardware Tools and Safety Precautions **UNIT-II** Computer Cables: • Various Types of Cables Used in Computer, Connectors and Ports used. • Connecting Cables from SMPS To Motherboard & Hard Disk. • Establishing Data Connection for Mother Board, Hard Disk, and Drivers. • Fixing Wires for Power Restart, Fixing Wires for HDD, LED, External USB & Audio Connections. • Cable Management Techniques. **UNIT-III** System Diagnostics, Troubleshooting and Maintenance: • Overview of the Troubleshooting Process and the Importance of Systematic Approach • Introduction To Hardware Diagnostic Tools (PC Toolkit) • Troubleshooting Common Hardware Problems in Keyboard, Mouse, Mouse, Printer, SMPS, Hard Disk, Motherboard, and Processor. • Maintenance of Keyboard, Mouse, Mouse, Printer, SMPS, Hard Disk, Motherboard, and Processor. • Breakdown and Breakthrough of various Hardware Issues. • Safety Precautions During Troubleshooting. **UNIT-IV** Window Installation: • Installing Drivers. • Installing and Updating Anti-Virus Software. • Hard Disk Partitioning (Creating, formatting & Deleting Partitions) Using DISKPART in Command Prompt. • Hard Disk Partitioning (Creating, formatting & Deleting Partitions) using Graphical User Interface.
11	**Suggested Resources for Reading** 1. Comdex Hardware and Networking Course Kit, Gupta, V., Dreamtech Press. 2. IBM-PC and Clones - Hardware Troubleshooting and Maintenance, Govindarajalu, B., Tata McGraw Hill. 3. Computer Hardware and Practical on Networking, Osheen Sharma and Shruti, Mohindra Publishing House. 4. Mastering PC Hardware & Network, Dr. Ajit Mittal, Dr. Ajay Rana, Khanna Book Publishing Company. 5. PC Hardware: A Beginner's Guide by Ron Gilster, McGraw-Hill Education. **Note**: Latest edition of text books may be used.

Semester II

<table>
<tr><th colspan="2">Course Title</th><th>Data Communication and Networks</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-201</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 4 marks each.
iv. The students are required to attempt one long question of 16 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Field Technician- Computing & Peripherals</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course aims to familiarize students with diverse network types, detailing their roles in computer environments. It provides a foundational understanding of network security concepts, emphasizing the importance of safeguarding infrastructure. Students will also learn switching techniques, IP address fundamentals and configuration.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Develop a comprehensive understanding of computer networks, analyze, and compare various network topologies and understand different network models.
CO2. Identify and describe different types of servers, understanding their roles and functionalities in network environments.
CO3. Understand transmission media, various networking protocols and the functionalities of network devices.
CO4. Develop a basic understanding of network security concepts, recognizing the importance of safeguarding network infrastructure.
CO5. Understanding IP addresses and its configuration.</td></tr>
<tr><td>9</td><td>Pedagogy of the</td><td>Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.</td></tr>
</table>

	Course Work	
10	**Course Content** **UNIT-I** Computer Network: Introduction, Characteristics, Uses of Computer Networks, Types of Networks Based on Area Coverage: LAN, WAN, MAN, Advantages and Disadvantages of Networks, Network Topologies: Bus Topology, Ring Topology, Star Topology, Tree Topology, Mesh Topology and Hybrid Topology. Network Models: Peer-To-Peer, Protocol, Client/Server Network and Hybrid Type Network. OSI Reference Model, TCP/IP Reference Model, Comparison of OSI with TCP/IP Model. Servers: Different Types of Servers. **UNIT - II** Data Communications: Analog Signal, Digital Signal, Analog Vs Digital Communication, Band Width Limitation, Data Rate of a Channel, Physical Layer: Transmission Media: Guided (Twisted-Pair, Coaxial and Optical Fiber) and Unguided (Radio, Microwave, and Infrared), Switching: Circuit Switching, Packet Switching, Message Switching. Telephone System, Modems. Modulation Techniques: Amplitude Modulation, Frequency Modulation, Phase Modulation. Multiplexing: Frequency Division, Time Division, Wave Length Division. **UNIT-III** Networking Protocols: Introduction, Domain Name System (DNS), DNS Name Space, DNS Servers, FTP, Telnet, DHCP, HTTP, TFTP, SSH, SFTP, SMTP, POP, IMAP, NTP. Network Devices: HUB, Switch, Repeater, Bridges, Routers, Gateway. IEEE Standards: IEEE Standards for LAN and WAN: IEEE Standard 802.3 and Ethernet, IEEE Standard 802.4 Token Bus, IEEE Standard 802.5 Token Ring, IEEE Standard 802.6 Distributed Queue Dual Bus, IEEE Standard 802.2 Logical Link Control. Introduction to Network Security. **UNIT-IV** IP Address: Introduction, IP Address Format, Overview of IP Address Classes: A, B, C, D, and E, Characteristics and Range of Each IP Address Class, Use Cases for Different IP Address Classes. Subnetting-Sub Network, Subnet Mask, Configuration of IP Address in PC, Differentiation Between IPv4 and IPv6. Access Points, Network Interface Card (NIC): Working Principle of NIC, MAC Address. User Management: Overview of User Account, Managing User Account and Password Security: Adding User, Setting and Changing Password, Changing Picture of User, Deleting the User, Setting Account Restriction for The User, Renaming a Local Account.
11	**Suggested Resources for Reading** 1. Computer Networks, Tanenbaum, Andrew, Fifth Edition, PHI. 2. Data Communication and Networking, Behrouz A. Forouzan, Fourth Edition, McGraw Hill Education. 3. Computer Today, S.K. Basandra, First Edition, Galgotia. 4. Data Communication System, Black, Ulysse, Third Edition, PHI. 5. Data and Computer Communications, Stalling, Ninth Edition, PHI. **Note:** Latest edition of text books may be used.	

<table>
<tr><th colspan="2">Course Title</th><th>Computing Software</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-202</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practicals
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4X 2 marks) = 8 marks
(4X8 marks) = 32 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Field Technician- Computing & Peripherals</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course covers essential skills in word processing, spreadsheet management, presentation creation, web browsing, and email communication. Students will learn document handling, formatting, and advanced features in Microsoft office applications. Additionally, they will acquire proficiency in web browsing, email etiquette, organization strategies, and customization options, enhancing their overall digital literacy.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Develop proficiency in using office suite applications, including word processing, spreadsheets, and presentation tools, for efficient document creation, data analysis, and presentation development.
CO2. Develop technological literacy by understanding and utilizing the latest features and tools available in office suite applications.
CO3. Students will be able to develop engaging presentations, mastering slide design, animations, and transitions, enabling effective communication through multimedia slideshows.
CO4. Understand basic communication tools, web surfing techniques, e-governance website usage, and understand email security settings, two-factor</td></tr>
</table>

		authentication, and safe practices for online communication.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.
10	**Course Content**	

UNIT-1

Word Processing Package: Opening, Saving and Closing an Existing Document, Renaming and Deleting Files, Using Styles and Templates: Introduction to Templates and Styles, Applying, Modifying and Creating New (Custom) Styles, using a Template to Create a Document, Creating a Template, editing a Template, Organizing Templates, Examples of Style Use, Changing Document Views, Moving Quickly Through a Document.

Working With Text: Select, Cut, Copy, Paste, Find and Replace, Inserting Special Characters, Setting Tab Stops and Indents, Checking Spelling and Grammar, Autocorrect, Using Built-In Language Tools, Word Completion, Autotext, Formatting Text: Using Styles, Formatting Paragraphs, Formatting Characters, Autoformatting, Creating Lists.

Formatting Pages: Using Layout Methods, Creating Headers and Footers, Numbering Pages, Changing Page Margins, Adding Comments to a Document, creating a Table of Contents, Creating Indexes and Bibliographies, Printing a Document, Using Mail Merge, Tracking Changes to a Document, Using Fields, Linking to Another Part of a Document, Using Master Documents, Creating Fill-In Forms.

UNIT-II

Spreadsheet Package: Worksheet Basics, Data Entry in Cells: Entry of Numbers, Text and Formulae, Moving Data in a Worksheet, Moving Around in a Worksheet, Selecting Data Range, Using the Interface (Toolbars, Menus), Editing Basics, Working with Workbooks, Cell Referencing. Formatting and Calculations: Using Auto Fill, Working with Formulae, Efficient Data Display with Data Formatting (Number Formatting, Date Formatting Etc.), Working with Ranges, Worksheet Printing.

Working with Graphs and Charts: Creating Embedded Chart Using Char Wizard, Sizing and Moving Parts, Updating Charts, Changing Chart Types, Chart Wizard, Adding Titles, Legends and Gridlines, Printing Charts, Database Management. Finding Records with Data Form, Adding/Deleting Records, Filtering Records in a Worksheet. Functions and Macros: Worksheet Creating Macros, Recording Macros, Running Macros, Assigning Macros to Buttons, Defining Macros from Scratch. Multiple Worksheets.

UNIT-III

Presentation Package: Introducing Power Point, Use of Power Point Presentation, Creating New Slides Saving, Opening, and Printing. Home Tab: New Slide, Layout, Reset, Delete, Setting Text Direction, Align Text, Convert to Smart Art, Drawing Options. Insert Tab: Table, Picture, Clipart, Photo Album, Smart Art, Shapes and Chart, Movie and Sound, Hyperlink and Action, Text Box, Word Art, Object.

Design Tab: Page Setup Options, Slide Orientation, Applying Various Themes, Selecting Background Style, and Formatting it. Animations Tab: Custom Animation for Entrance, Exit and Emphasis, Applying Slide Transition, Setting Transition Speed and Sound, Animation on Rehears Timing. Slide Show &View Tab: Start Slide Show Options, Setup Options. View Tab: Presentation Views, Colours, and Window Option.

UNIT-IV

Web Browsing: Popular Web Browsing Softwares, Search Engines, Search for Content Accessing Web Browser, Using Favorites Folder, Downloading Web Pages, Printing Web Pages, Understanding URL, Surfing the Web, Using E-Governance Website.

Communication: Overview of E-Mail and its Evolution, Creating and Customizing E-Mail Account,

	Understanding E-Mail User Interface and Layout, Basic Email Etiquette, Essential Email Features, Composing, Sending, and Receiving Emails, Attachments and File Sharing, Labels and Filters for Organizing Emails, Advanced Search Techniques. Organization Strategies: Inbox Organization and Management, Priority Inbox and Tabbed Inbox Features, Using Folders (Labels) Effectively, Archiving and Deleting Emails, Advanced Settings, and Customization: Customizing E-Mail Settings for Personal Preferences, Signature Creation and Customization, Vacation Responders and Out-of-office Settings, Security Settings and Two-Factor Authentication, Mobile and offline Access: Configuring E-Mail on Mobile Devices, offline Access, and Synchronization.
11	**Suggested Resources for Reading** 1. Microsoft Windows Networking Essentials 1st Edition, Darril Gibson, Sybex. 2. Computer Fundamentals, Sinha P.K. & Sinha Priti, BPB Publications. 3. DOS 6.2 Quick Reference, Mathur Rajiv, Galgotia. 4. Introduction to Computers with MS - Office, Leon, TMH Publication 5. Using the Internet, Kasser, Barbara, PHI, 4th ed., New Delhi. 6. Using the World Wide Web, Wall, David A. & Others, PHI, 2nd ed., New Delhi. **Note**: Latest edition of text books may be used.

BHN-202P (Practical Based on BHN-202)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

<table>
<tr><th colspan="2">Course Title</th><th>Network Configuration</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-203</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>120 hours of practicals
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>-</td><td>-</td><td>80</td><td>20</td></tr>
</table>
i. The exam of this course will be of 3hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
Instructions for paper setters and candidates:
i. Examiner will set total of four questions covering the whole syllabus.
ii. Student will attempt any three questions.
iii. Each question carries 10 marks.
iv. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Field Technician- Computing & Peripherals</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course aims to provide a comprehensive understanding of networking tools, connectors, cables, and testing instruments. Participants will learn to establish LAN connections, configure internet connectivity, and implement Wi-Fi setups. The course emphasizes hands-on skills in cable termination, network testing, and troubleshooting.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Proficiently utilize networking tools like crimping and punching tools.
CO2. Demonstrate expertise in working with various connectors, including RJ-45 and Fiber LC.
CO3. Construct and troubleshoot LAN connections using UTP cables, switches, and testing tools.
CO4. Execute networking commands for configuration and diagnostics.
CO5. Configure and secure internet and Wi-Fi connectivity, demonstrating knowledge of different internet connection types.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>10</td><td colspan="2">Course Content

UNIT-I
Working Using Various Networking Tools
• Crimping Tool, Punching Tool, Punch Down Tool, Multi Punch Tool, RJ45 Punch Down Tool, Computer Punch Tool
• Cable Cutter, I/O Device, I/O Face Plate, I/O Box, Lan Cards</td></tr>
</table>

	Working Using Various Connectors • RJ-11, RJ-45, F-Type, ST (Straight Tip) • SC (Subscriber Connector or Standard Connector) • Fiber LC (Local Connector) • MT- RJ (Mechanical Transfer Registered Jack) • USB (Universal Serial Bus) **UNIT-II** Networking Cables, Patch Cords and Testing Tools: • Working of Different Cables: Coaxial Cable, Unshielded Twisted Pair (UTP), Types of UTP Cable, Shielded Twisted Pair Cable, Fiber Optics. • Patch Cords: UTP Patch Cords, Fiber Patch Cords. Cable Manager, Cable Tag, Cable Stripper, Scissors. • Testing Tools: LAN Testers, Application Tools- IP Scanner, Test Network Speed, Bandwidth Monitor 2 Lite Edition, Netbrute Scanner. **UNIT-III** Establishing Lan Connection: • Patching of LAN Cable, Strip Cable End, Untwist Wire Ends. • Arrange Wires, Trim Wires to Size, Attach Connector, Check, Crimp, Test. • Setting Up of LAN Using UTP Cables, Switch, LAN Tester, Switch Rack, Patch Panel, Cable Manager, Cable Tagging, I/O Box, I/O Face Plate, Patch Cords, Connectors, Terminals. Networking Commands: • ping, ipconfig, fstat, iproute, route, netstat, ipconfig/release, ipconfig/renew. **Unit-IV** Internet Connectivity: • Different Types of Internet Connection (Wireless, Mobile, Hotspot, Dial-Up, Broadband, DSL, Cable, Satellite, ISDN.) Wi-Fi Connectivity: Basic Configuration of Wi-Fi Device or Access Point, IP Address Configuration, DNS Address, Security and Password, Admin Login and Password, MAC Filtering and Monitoring.
11	**Suggested Resources for Reading** 1. Microsoft Windows Networking Essentials 1st Edition, Darril Gibson, Sybex. 2. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education. 3. Cisco Networking Essentials, Troy Mcmillan, Sybex. 4. Comdex Hardware and Networking Course Kit, Vikas Gupta, 1st Edition, Dreamtech Press. 5. Computer Hardware and Practical on Networking, Osheen Sharma and Shruti, Mohindra Publishing House. 6. Wireless Communication: Principle and Practices, Theodore S. Rappaport, Pearson Publication. **Note**: Latest edition of text books may be used.

Semester III

<table>
<tr><th colspan="2">Course Title</th><th>Basics of Routing and Switching</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-301</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 4 marks each.
iv. The students are required to attempt one long question of 16 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, it enables the students to enhance career development skill in the field of networking.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Identify different types of routers, describe their functions, modes of operation and configure basic router settings.
CO2. Differentiate between static and dynamic routing, configure static routes.
CO3. Understand and implement dynamic routing protocols and verify their configurations.
CO4. Understand Switch Functions and its configuration.
CO5. Configure and Manage VLANs.</td></tr>
<tr><td>9</td><td>Pedagogy of the</td><td>Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.</td></tr>
</table>

	Course Work	
10	**Course Content** **UNIT-I** Router: Introduction, Components of routers, Types of Routers: Broadband Routers, Wireless Routers, Edge Router, Subscriber Edge Router, Inter-provider Border Router, Core Router, Wired and Wireless Routers. Functions of routers, Router modes, Configuration of Router: names, passwords, password encryption, interfaces name, creating a login banner, saving configuration and erasing configuration. **UNIT - II** Routing: Introduction, Types of routing: static and dynamic routing, Static routing: Configuring a static route, permanent keyword, static routes, and administrative distance, configuring default route. Dynamic routing: Dynamic Routing Protocols: RIPv1, RIPv2, IGRP, EIGRP and OSPF. Delivery Semantics: unicast, anycast, multicast, broadcast, geocast. Configuring and Verifying RIP, IGRP, OSPF, EIGRP. **UNIT-III** Switch: Introduction, roles of switches in network, types of switches: managed and unmanaged switch. Working of switch in different layers. Configuration of Switch: Command Mode, Setting Host Names, Setting Passwords, Setting IP Addresses and Default Gateways, Setting Operational Speed, Switch Port Security, Sticky MAC Addresses. **UNIT-IV** Switching: Introduction, Switching Services, Bridging vs LAN Switching, Switch Functions at Layer2, Definition of Spanning Tree Protocol (STP), Operations of STP, STP Port States, Definition of VLAN, Typing of VLAN: Static and Dynamic, VLAN Identification Methods, VLAN Trunking Protocol (VTP), VTP Modes of Operation, Routing between VLANs, Configuring Inter-VLAN Routing.
11	**Suggested Resources for Reading** 1. Lacoste, CCNP Routing and Switching TSHOOT 300 - 135: Official Cert Guide, Pearson India 2. Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd 3. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education 4. Vikas Gupta, Comdex Hardware and Networking Course Kit, 1st Edition, Dreamtech Press. 5. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media. 6. Networking: The Complete Reference 1st Edition, Mcgraw Hill Education

<table>
<tr><th colspan="2">Course Title</th><th>Programming Using C Language</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-302</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practicals
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4X 2 marks) = 8 marks
(4X8 marks) = 32 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>The objective of this course is to make the student understand programming language concepts, mainly control structures, reading a set of data, stepwise refinement, function and arrays. After completion of this course, the student is expected to analyze the real-life problem and write programs in ‘C’ language to solve problems. The main emphasis of the course is on problem solving aspect.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Understand the programming concepts.
CO2. Learn the C programming in practical usage.
CO3. Solve programming problems using C programming language.
CO4. Gain the in-depth knowledge of strings, arrays, functions and file handling etc.
CO5. Develop applications based on programming concepts.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.</td></tr>
</table>

10	**Course Content** **UNIT-I** Programming Process: Steps in developing a program, Algorithm development, Flowchart, Testing and Debugging. Fundamentals of C Languages: History of C, Character Set, Identifiers, Keywords, Constants, Types of C Constants, Variables, Data Types. Operators and Expressions: C Instructions, Arithmetic operators, Relational operators, Logical operators, Assignment Operators, Hierarchy of Operations, Structure of a C program, Compilation and Execution. **UNIT-II** Decision Control Structure: IF-statement, IF-Else and Nested IF-Else, The else if Clause. Loop Control Structure: While and do-while, for loop and Nested for loop, Case Control Structure: Decision using switch, goto statement. Functions: Library functions and user defined functions, Global and Local variables, Function Declaration, Calling and definition of function, Methods of parameter passing to functions, recursion. **UNIT-III** Arrays: Introduction, Array declaration, accessing values in an array, initializing values in an array, Single and Two-Dimensional Arrays, initializing a 2-Dimensional Array, Memory Map of a 2-Dimensional Array, Arrays of characters, Insertion and deletion operations. Pointers: Pointer declaration, Address operator "&", Indirection operator "*", simple implementation of pointers. **UNIT-IV** String Manipulation in C: Declaring and Initializing string variables, Reading and writing strings, String functions (strlen(), strcpy(), strcmp(), strcat()). Structures and Unions: Declaration of structures, Structure Initialization, Accessing structure members, Union. Files in C: Introduction, Opening and Closing files, Basic I/O operation on files.
11	**Suggested Resources for Reading** 1. Yashavant P. Kanetkar: Let us C, BPB Publications, New Delhi. 2. Salaria, R.S.: Test Your Skills in C, Salaria Publications, New Delhi. 3. C. Balaguruswami: Programming with C Language, Tata McGraw Hill, New Delhi. 4. Byron S. Gottfried: Programming in C, McGraw Hills Publishers, New York. 5. M.T. Somashekara: Programming in C, Prentice Hall of India. **Note**: Latest edition of text books may be used.

BHN-302P (Practical Based on BHN-302)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

<table>
<tr><th colspan="2">Course Title</th><th>Routing Configuration</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-303</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>120 hours of practicals
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>-</td><td>-</td><td>80</td><td>20</td></tr>
</table>
The exam of this course will be of 3hours duration.
i. There will be no objective type questions.
ii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iii. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
Instructions for paper setters and candidates:
i. Examiner will set total of four questions covering the whole syllabus.
ii. Student will attempt any three questions.
iii. Each question carries 10 marks.
iv. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>Routing technology provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Describe basic routing concepts, configure and verify basic router settings using CLI.
CO2. Differentiate between static and dynamic routing, link state and distance vector protocols, and analyze key routing concepts.
CO3. Configure and verify inter-VLAN routing.
CO4. Test network connectivity and troubleshoot configurations.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>10</td><td colspan="2">Course Content

UNIT-I
Functionality of Router and Switch: Describe basic routing concepts: packet forwarding, router lookup process, Process Switching/Fast Switching/CEF.
Configure and verify utilizing the CLI to set basic Router configuration: hostname, local user and password, enable secret password, console and VTY logins, exec-timeout, service password encryption, Interface IP Address, loopback, banner, motd, copy run start.

UNIT-II
Configure and verify operation status of a device interface: Serial, Ethernet, Verify router configuration and network connectivity using, ping, extended, traceroute, telnet, SSH, shcdp neighbors. The boot</td></tr>
</table>

	process of Cisco IOS routers, Configure and verify operation status of a device interface, both serial and Ethernet, Verify router configuration and network connectivity. IP Routing Technologies: Configure and verify routing configuration for a static or default route given specific routing requirements. **UNIT-III** Introduction to routing and routing protocols, Static versus Dynamic Routing, Link state versus Distance Vector, next hop, IP routing table, Passive Interfaces and its working, Admin Distance, split horizon, metric. Configure and Verify OSPF: Benefit of single area, Configure OSPv2, Configure OSPv3, Router ID, Passive Interface, Multi-area OSPF, LSA types and purpose. **Unit-IV** Configure and verify inter VLAN routing (Router on a stick), sub interfaces, upstream routing, encapsulation, Configure SVI interfaces, Manage Cisco IOS Files, Boot Preferences, Cisco IOS Images (15), Licensing, Show license, Change license, Configure and verify EIGRP (single AS), Feasible Distance/Feasible Successors/Administrative distance, Feasibility condition, Metric composition, Router ID, auto summary, Path Selection, Load Balancing, Unequal, Equal.
11	**Suggested Resources for Reading** 1. Lacoste, CCNP Routing and Switching TSHOOT 300-135: Official Cert Guide, Pearson India 2. Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd 3. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education 4. Mcmillan,Cisco Networking Essentials, Author: John Wiley & Sons 5. Crypto and Network Security, 3rd Edition, Mcgraw Hill Education 6. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media **Note**: Latest edition of text books may be used.

Semester IV

<table>
<tr><th colspan="2">Course Title</th><th>Digital Electronics and Computer Organisation</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-401</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 4 marks each.
iv. The students are required to attempt one long question of 16 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course is an introduction to the basic principles of digital electronics and computer organisation. At the conclusion of this course, the student will be able to quantitatively identify the fundamentals of computers, including number systems, logic gates, logic, arithmetic subsystems, and integrated circuits. The student will be able to analyze and design simple logic circuits and draw logic diagrams.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Understand significance of number systems and its conversions.
CO2. Illustrate knowledge on design of various combinational circuits.
CO3. Illustrate the concept of sequential logic design, analyze the operation of flip-flops, registers, and counters
CO4. Discuss the basic structure and organization of computers.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.</td></tr>
</table>

10	**Course Content** **UNIT-I** Introduction to Digital Electronics: Digital vs Analog Systems, Analog vs digital signals, advantages of digital systems, Number Systems (Binary, Decimal, Octal, Hexadecimal) and Conversion between different number systems and Binary Arithmetic (Binary addition, subtraction). Logic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR), Truth tables and symbolic representations. Introduction to Computer Systems: Basic Computer Architecture, Stored program concept and Von Neumann architecture, Basic Operations in a Computer (CPU operations: Fetch-Decode-Execute cycle), Arithmetic and Logic Unit (ALU) and Control Unit (CU). **UNIT - II** Combinational Circuits: Half adder, Full adder, Subtractors, Multiplexers, Demultiplexers, Encoders, Decoders. CPU Design: Components of the CPU: ALU, Registers and Types of registers (Accumulator, Program Counter, Instruction Register), Internal communication: Buses, data lines, address lines, Basic understanding of the Instruction Set Architecture (ISA), addressing modes, instruction pipelining**.** **UNIT-III** Instruction Types: Data transfer, Arithmetic, Logic, Branching, and Control instructions. Example of a basic instruction set for simple processors. Sequential Circuit: Flip-Flops (SR, D Flip-Flop, T Flip-Flop, JK Flip-Flop) and their truth tables, race around condition, Master Slave flip flop, Shift registers: SIPO, PISO, PIPO, SIPO, Bi-directional; Counters: ripple & synchronous counters - up / down. **UNIT-IV** Memory Organization and Architecture: Memory Hierarchy, Cache memory (levels of cache, hit/miss rate), RAM, ROM, Virtual Memory. Memory Access and Optimization: Memory Access Techniques (Direct Memory Access (DMA), Memory-Mapped I/O), Cache Misses and Optimization. Processor Performance and Optimization: Processor Performance Metrics, Performance Optimizations.
11	**Suggested Resources for Reading** 1. M. Morris Mano. Computer System Architecture, Prentice Hall International 3 rd ed., (1993). 2. P. Pal Choudhar. Computer Organisation and Design, Prentice Hall of India, (1994). 3. B. Govindarajalu, IBM-PC and Clones - Hardware Troubleshooting and Maintenance, Tata - McGraw Hill, (1994).

<table>
<tr><th colspan="2">Course Title</th><th>Network Administration</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-402</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practicals
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4X 2 marks) = 8 marks
(4X8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>The goal of this course is to train students to an exceptionally high standards in network planning, implementation, administration and support.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Perform IOS Software and Configuration Management tasks.
CO2. Understand and Implement Password Recovery Procedures.
CO3. Apply Basic Troubleshooting Techniques and Enhance Network Security and Performance.
CO4. Utilize Security Device Manager (SDM) for Network Configuration.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.</td></tr>
<tr><td>10</td><td colspan="2">Course Content
UNIT-I
Backing UP and Restoring IOS Software and Configuration: Boot system commands, IOS file system, backing up configuration to a TFTP server, restoring configuration to a TFTP server, backing up the IOS software to a TFTP server, restoring/upgrading the IOS software from a TFTP server, restoring IOS software from ROM Monitor mode using Xmodem.</td></tr>
</table>

	UNIT-II Password-Recovery Procedures and configuration Register: The configuration register: A visual representation, bit means, the boot field, console terminal baud rate setting. Console line speed: CLI & ROM Monitor mode, password recovery procedures for router, password recovery procedures for switches. **UNIT-III** Basic troubleshooting: viewing routing table, determining the gateway of last resort, last routing update, OSI layer 3 testing, OSI layer 7 testing, interpreting the show interface command, using CDP to troubleshooting, show controllers command debug commands, time stamps, Operating system IP verification commands, the ip http server command, netstat command. **UNIT-IV** Security Device Manager: SDM connecting with CLI and GUI, SDM Express Wizard with no CLI preconfiguration, resetting the router to factory defaults, SDM user interfaces, configuration routing using SDM, SDM monitor mode, SDM to configuration a router as a DHCP server & client, SDM configuration NAT/PAT.
11	**Suggested Resources for Reading** 1. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education 2. Cisco Networking Essentials, Author: McMillan, troy author; John Wiley & Sons 3. Vikas Gupta, Comdex Hardware and Networking Course Kit, 1st Edition, Dreamtech Press 4. Crypto and Network Security, 3rd Edition, Mcgraw Hill Education 5. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media 6. Networking: The Complete Reference 1st Edition, Mcgraw Hill Education **Note**: Latest edition of text books may be used.
BHN-402P (Practical Based on BHN-402) Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation. 1. Examiner will set total of four questions covering the whole syllabus. 2. Student will attempt any three questions. 3. Each question carries 5 marks. 4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.	

<table>
<tr><th colspan="2">Course Title</th><th>DOS and Windows Operating System</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-403</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practicals
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4X 2 marks) = 8 marks
(4X8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course enables students to understand the concepts of Operating System. After the completion of this paper, student will be able to manage various processes, files and folders on the disk and desktop with effective outcome.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Demonstrate proficiency in DOS.
CO2. Understanding Windows operating systems and performing basic program management tasks.
CO3. Configure system properties, manage devices through the device manager, and use troubleshooting tools.
CO4. Utilize various tools for maintaining and optimizing storage devices, and managing folders and files.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>10</td><td colspan="2">Course Content</td></tr>
</table>

	UNIT-I DOS: Versions of DOS: Booting sequence; Warm and Cold reboot; Concept of File and directory, Redirecting command input and output pipes, Wildcard characters, Types of DOS commands: Internal and External; Internal Commands: DIR, MD, CD , CLS, COPY, DATE, DEL, PATH, PROMPT, REN, RD, TIME, TYPE, VER, VOL; External Commands: XCOPY, ATTRIB, BACKUP, RESTORE, FIND, SYS , FORMAT, CHKDSK, DISKCOPY, LABEL, MOVE, TREE, DELTREE, DEFRAG, SCA NDISK, UNDELETE. Batch Files: Introduction to simple batch files; Introduction to CONFIG.SYS and AUTOEXEC.BAT files. **UNIT-II** Windows Operating System: features, history, hardware requirements, and installation of Windows Operating System. Graphics interface: Benefits, Screen attributes: icons and bars, Accessories of the Windows program. Program management: Creating, saving and editing documents, Concurrent execution of programs, sharing data between programs. Folder and file management: Working with files, Naming files, Navigate to Folders with Windows Explorer, Copying and moving files, deleting files, managing folders, Creating, Viewing, Expanding and collapsing, Backing up and restoring files. **UNIT-III** Components of Windows: format of a window, moving windows, resizing windows, minimizing and maximizing windows. Control panel: Customizing screens, Screen colours, Patterns, Spacing icons, selecting time/date, setting the Sound. Creating Short cuts, Notepad, System restore. Customizing printing, changing the print queue, configuring the printer, Adding printers. Working with fonts: changing, removing, adding, customizing mouse and keyboard use. **UNIT-IV** System properties and the device manager Management tools, DOS sessions, Explorer, Memory configuration, Safe mode Install and uninstall applications, Setup/troubleshooting issues. Maintaining and optimizing disks: Disk Cleanup, Disk defragmenter, Compressing and uncompressing folders and files. Encrypting and decrypting folders and files.
11	**Suggested Resources for Reading** 1. Jamsa, Kris A., Rescued by Windows 3.1, Galgotia. 2. Basandra, S.K., Computers Today, Galgotia. 3. Mathur Rajiv, Learning Word 6 for Windows Step-by-Step, Galgotia. 4. Mathur Rajiv, Learning Excel 5 for Windows Step-by-Step, Galgotia. **Note**: Latest edition of text books may be used.
	BHN-403P (Practical Based on BHN-403) Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation. 1. Examiner will set total of four questions covering the whole syllabus. 2. Student will attempt any three questions. 3. Each question carries 5 marks. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

<table>
<tr><th colspan="2">Course Title</th><th>Switching Configuration</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-404</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>120 hours of practicals
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>-</td><td>-</td><td>80</td><td>20</td></tr>
</table>
The exam of this course will be of 3hours duration.
i. There will be no objective type questions.
ii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iii. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
Instructions for paper setters and candidates:
i. Examiner will set total of four questions covering the whole syllabus.
ii. Student will attempt any three questions.
iii. Each question carries 10 marks.
iv. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>Switching technology provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Gain proficiency in configuring and verifying DHCP, ACLs, and NAT for network operations, addressing various network requirements and enhancing security.
CO2. Configure and verify NTP, Syslog, and SNMP to ensure synchronized time, event logging, and effective network monitoring.
CO3. Configure and verify switch operations and understand switching methods and concepts.
CO4. Configure enhanced switching technologies, port security features, secure unused ports and optimize VLAN configurations to strengthen network security.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>10</td><td colspan="2">Course Content

UNIT-I
IP Services: Configure and verify DHCP (IOS Router), Configuring router interfaces to use DHCP, DHCP options (basic overview and functionality), Excluded addresses, Lease time, Describe the types, features, and applications of ACLs, standard (editing and sequence numbers) extended, named, numbered, Log option. Configure and verify ACLs in a network environment, named, numbered, Log</td></tr>
</table>

	option, Identify the basic operation of NAT, purpose, pool, static, 1 to 1, overloading, source addressing, one way NAT, Configure and verify NAT for given network requirements. **UNIT-II** Configure and verify NTP as a client. Recognize High availability (FHRP), VRRP, HSRP, GLBP, and Configure and verify syslog, Utilize Syslog Output, Describe SNMP v2 and v3. LAN Switching Technologies: Determine the technology and media access control method for Ethernet networks, Identify basic switching concepts and the operation of Cisco switches: Collision Domains, Broadcast Domains, Ways to switch, Store, Forward, Cut through, CAM Table. **UNIT-III** Configure and verify initial switch configuration including remote access management: Hostname, management IP address, IP default-gateway, local user and password, enable secret password Console and VTY logins, exec-timeout service password encryption, copy run start, verify network status and switch operation using basic utilities such as: ping, telnet, SSH. Describe how VLANs create logically separate networks and the need for routing between them, network segmentation and basic traffic management concepts, Configure and verify VLANs. **Unit-IV** Configure and verify trunking on Cisco switches: DTP (topic), auto-negotiation, Identify enhanced switching technologies: RSTP, PVSTP, Ether channels, Configure and verify PVSTP operation: Describe root bridge election, Spanning tree mode. Configure and verify Switch Port Security: Sticky mac, MAC address limitation, static/dynamic, violation modes, err disable, shutdown, protect restrict, shutdown unused ports, err disable recovery, assign unused ports in unused VLANs, putting Native VLAN to other than VLAN 1.
11	**Suggested Resources for Reading** 1. Lacoste, CCNP Routing and Switching TSHOOT 300 - 135: Official Cert Guide, Pearson India 2. Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd 3. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education 4. Mcmillan,Cisco Networking Essentials, Author: John Wiley & Sons 5. Crypto and Network Security, 3rd Edition, Mcgraw Hill Education 6. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media **Note**: Latest edition of text books may be used.

Semester-V

<table>
<tr><th colspan="2">Course Title</th><th>Information System Design</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-501</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 4 marks each.
iv. The students are required to attempt one long question of 16 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Administrator</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>To teach the students about the various aspects of Information Systems to be developed their analysis and design. The motive of this syllabus is to make learners aware about pre requisite of Information System and associated paradigms. After completing this course student will be able to be analyse and design information systems.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. To understand System Development Life Cycle and System Analyst role.
CO2. Understanding of SRS, Flow diagrams, and other necessary tools.
CO3. Understand feasibility study, testing and quality assurance of projects.
CO4. To know system implementation, maintenance, and installation of projects.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.</td></tr>
</table>

10	**Course Content** **UNIT - I** Systems Concepts and Information Systems Environment: Definition and characteristics of a system. Elements of a system Environment: Boundaries and interface. Types of systems: Physical or Abstract Systems, Open and Closed System, Man - made information systems. Types of System Models. System Development Life Cycle: Introduction to various phases-Recognition of Need, Feasibility Study, Analysis, Design, Implementation, Post- Implementation and Maintenance. The Role of System Analyst: Skills of a System Analyst, Duties of the System Analyst. **UNIT - II** System Planning and the Initial Investigation: Bases for planning in system analysis, Initial investigation, determining the user's information requirements, Problem definition and Project Initiation, Background Analysis, Fact Finding, Fact Analysis, Determination of Feasibility. Information Gathering: Introduction, Information Gathering tools: Review of Literature, Procedures, and forms. On -site observation. Interviews and questionnaires. Tools of Structured Analysis: SRS features and structure, Various tools of structured analysis: Data flow diagram (DFD), Data Dictionary, Decision tree and structured English, Decision table, Pros and cons of each tool, ER-Diagrams **UNIT - III** Feasibility Study: System Performance-statement of Constraints, Identification of Specific System Objectives, description of Outputs. Feasibility Study - Feasibility considerations, Steps in feasibility analysis. Feasibility Report. System Design: The Process of Design-Logical and Physical Design, Design methodologies: Structured design, Functional Decomposition. System Testing: Testing, System testing, test phases, types of System Testing, WBT and BBT techniques, OO testing. Quality Assurance: Quality assurance and its goals in its system life cycle, Levels of quality assurance, Trends in testing. **UNIT - IV** Implementation and Software Maintenance: Introduction, Conversion- Activity network for Conversion, File Conversion, User Training: Elements of user Training Post implementation review. Software Maintenance - Primary activities of a Maintenance Procedure, Reducing Maintenance Costs. Types of Software Maintenance. Hardware and Software Selection: Types of Software, Procedure for Hardware/Software selection: Major phases in selection, Evaluation and Validation, Vendor Selection, Post - Installation Review. Software selection- Criteria for Software Selection, the evaluation process.
11	**Suggested Resources for Reading** 1. WS Jawadekar and SS Dubey, Management Information System, McGraw Hill; Sixth edition 2020 2. Alan Dennis, Barbara Haley Wixom, Roberta M. Roth., Systems Analysis and Design, Edition 2018, John Wiley & Sons Inc. 3. Hardgrave Bill C., Siau Keng, Chiang Roger H.L., Systems Analysis and Design: Techniques, Methodologies, Approaches and Architectures, Edition 2017, M.E. Sharpe Publications. 4. E.M. Awad: Systems Analysis and Design, Edition 2015, Galgotia Publications **Note**: Latest edition of text books may be used.

<table>
<tr><th colspan="2">Course Title</th><th>Wireless Communication and Networking</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-502</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 4 marks each.
iv. The students are required to attempt one long question of 16 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Administrator</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>To introduce students to wireless communication basics, cellular system concepts, wireless networking, and modern wireless technologies so they can understand how mobile, Wi-Fi, and broadband wireless systems work.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1: Understand the basic concepts, evolution, and types of wireless communication systems.
CO2: Explain cellular system design fundamentals such as frequency reuse, handoff, interference, and multiple access techniques.
CO3: Describe wireless networking, routing, wireless data services, and technologies.
CO4: Understand wireless home networking, HyperLAN, ATM, and the principles</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.</td></tr>
<tr><td>10</td><td colspan="2">UNIT-I
Wireless communication systems: Introduction-Evaluation of mobile radio communications, wireless communication systems example, paging systems, cordless telephone systems, various wireless systems compression. Mobile wireless communication systems: 2nd generation cellular networks, 3rd generation wireless networks, wireless in local loop, wireless LAN, Bluetooth and personal area networks (PAN).</td></tr>
</table>

	UNIT-II Cellular system design fundamentals: Allocation of spectrum, basic cellular system, frequency reuse, channel assignment strategies, handoff strategies, interference and system capacity, trucking and grade off service, improving coverage and capacity, cell splitting. Multiple access technique for wireless communications: introduction to multiple accesses, FDMA, TDMA, spread spectrum multiple access, SDMA, packet radio, capacity of cellular systems. **UNIT-III** Wireless Networking: Difference between wireless and fixed telephone networks, development of wireless networks, fixed network transmission hierarchy, traffic routing in wireless networks, wireless data services, common channel signaling. Wireless WAN: mechanism to support at mobile environment, communication in the infrastructure, iIS-95 CDMA forward channel, IS-95 CDMA risers' channel, packet and frame formats in IS-95, IMT -20000, forward channel in W-CDMA and CDMA 2000, reverse channels in W-CDMA and CDMA -2000 GPRS and higher data rates, short messaging service in GPRS mobile application protocols. **UNIT-IV** Wireless land: Historical overviews of the land industry, evolution of the wan industry, wireless home networking IEEE 802.11 the PHY layer, Mac layer wireless ATM, Hyperlink, Hyper Lan-2 Orthogonal frequency division multiplexing: basic principles of orthogonality single versus multi-channel systems, OFDM block diagram, OFDM signal mathematical representation.
11	**Suggested Resources for Reading** 1. Wireless Communications, Principles, Practice - Theodore, S. Rappaport, PHI, 2nd Edn. 2. Wireless Communication and Networking - William Stallings, PHI. 3. Wireless Digital Communications - KamiloFeher, PHI. 4. Principles of Wireless Networks - KavehPahLaven and P. Krishna Murthy, Pearson Education. 5. Wireless Communications - Andreaws F. Molisch, Wiley India. 6. Introduction to Wireless and Mobile Systems - Dharma PrakashAgarwal, Qing-AnZeng, Thomson 2nd Edition. **Note**: Latest edition of text books may be used.

<table>
<tr><th colspan="2">Course Title</th><th>Linux Operating System</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-503</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practical
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4X 2 marks) = 8 marks
(4X8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Administrator</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>To introduce students to the fundamentals of operating systems and Linux, enabling them to understand system architecture, manage processes, and perform basic administration tasks on Linux-based systems. This course introduces students to operating system concepts with a focus on Linux, covering OS types, functions, and architecture. It develops practical skills in using Linux commands, managing files, processes, and communication tools.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Explain the fundamentals, functions, and types of operating systems, and describe the architecture and file system structure of the Linux operating system.
CO2. Use essential Linux commands for file handling, directory management, disk operations, filters, pipes, process management.
CO3. Create and edit files using the VI editor, apply various editing operations, and use Linux communication tools for messaging, mailing, and remote access.
CO4. Perform basic system administration tasks including user management, file permissions, Linux installation, startup, and shutdown procedures.</td></tr>
</table>

9	**Pedagogy of the Course Work**	Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.
10	**Course Content** **UNIT - I** Introduction to Operating Systems: Definition, need, functions, Types of Operating System: Batch Operating System, Time-Sharing (Multitasking), Distributed Operating System, Network Operating System (NOS), Real-Time Operating System (RTOS), Mobile Operating System, Multiprocessing OS, Multiuser OS, Embedded Operating System. Linux Operating System: features, Basic Architecture, features of Kernel and Shell. Linux File system - Boot block, super block, Inode table, data blocks, How Linux kernel access files, Linux standard file system. Structure of file system. **UNIT -II** Essential Linux commands - Commands for files and directories: cat, cd, ls, cp, md, rm, mkdir, rmdir, pwd, file, more, less, file comparisons - cmp&comm, View files, disk related commands, checking disk free spaces, chmod with its options, cal,date,who,tty, lp,stty. Filters and pipes: head, tail, wc, pr, cut, paste, sort, uniqe, grep, egrep, fgrep, tee. The process: shell process, parent and children, process status, system process, multiple jobs in background and foreground, changing process priority with nice, premature termination of process, Mathematical commands- bc, expr, factor, units **UNIT -III** Creating and editing files with VI editor with their command options, Operators, text deletion, text movement, changing text, yanking text, filtering text, the ex-mode, moving text from one file to another. Communication: The bulletin board -news, write, mesg, talk, mail, elm, pine, finger, vacation and connecting to remote machine. **UNIT -IV** System administration Common administrative tasks, identifying administrative files - configuration and log files, Role of system administrator, managing user accounts-adding & deleting users, changing permissions and ownerships, Installation of Linux system- Linux Installation requirement, complete Procedure steps, Partitioning the Hard drive, System startup and shut-down process, init and run levels. File system mounting, lpstat, backup strategy, installing software on Linux.	
11	Suggested Resources for Reading 1. Peterson, James, L. and Silberschatz, A.: Operating System Concepts, Wiley Publ. Comp, 1985. 2. Milenkovic, M.: Operating System - Concepts and Design, McGraw Hill, International Editions, 1987. 3. Joshi, R.C.: Operating Systems, Wiley India Pvt. Ltd. 4. Parker, Tim: Linux Unleashed, Latest Edition, Techmedia. 5. Stones, Richard and Mathew Neil: Beginning Linux Programming, 3rd Edition, Wrox **Note**: Latest edition of text books may be used.	
	BHN-503P (Practical Based on BHN-503) Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation. 1. Examiner will set total of four questions covering the whole syllabus. 2. Student will attempt any three questions. 3. Each question carries 5 marks. Practical file carries 5 marks and 20 marks for practical work and viva-voce.	

<table>
<tr><th colspan="2">Course Title</th><th>Web Designing</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-504</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>120 hours of practical
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>-</td><td>-</td><td>80</td><td>20</td></tr>
</table>
The practical exam of this course will be of 3hours duration.
i. There will be no objective type questions.
ii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iii. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
Instructions for practical paper setters and candidates:
i. Examiner will set total of four questions covering the whole syllabus.
ii. Student will attempt any three questions.
iii. Each question carries 10 marks.
iv. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Administrator</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>To enable students to design and develop basic webpages by using HTML for structure, CSS for styling, and JavaScript for adding interactivity. Students will learn how to apply essential web technologies to build simple, functional websites suitable for real-world applications.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Create structured webpages using HTML elements, tags, forms and multimedia components to build functional website content.
CO2. Apply CSS for styling, layout design, responsive structure and visual formatting to enhance webpage appearance.
CO3. Use JavaScript to implement basic interactivity, handle user events and perform simple form validations.
CO4. Develop complete, organized and user-friendly webpages by integrating HTML, CSS and JavaScript effectively.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>10</td><td colspan="2">Course Content
UNIT-I
Introduction to Web, Internet & HTML Basics
• Introduction to the internet, WWW, websites, web pages and web servers
• How the web works: URLs, HTTP/HTTPS, domain names, hosting
• Types of websites: static, dynamic, responsive, e-commerce, portals
• Introduction to HTML and webpage structure
• HTML tags, elements, attributes and nesting rules
• Creating a basic webpage using <html>, <head>, <body></td></tr>
</table>

	• Text formatting: headings, paragraphs, line breaks, bold, italic, underline • Adding images, hyperlinks, email links and anchor tags • Creating lists: ordered, unordered lists • Building tables with rows, columns and formatting, merging cells, captions, table styling **UNIT-II** **Page Structure & Forms** • Multimedia embedding: audio, video, YouTube, maps, iframes • HTML forms: purpose, syntax and structure • Form controls: input box, password box, radio buttons, checkboxes • Dropdown menus, list boxes, text areas, file upload control • Buttons: submit, reset, image button • Form attributes: action, method, target **Unit-III** **Cascading Style Sheets (CSS)** • What is CSS • Ways to apply CSS: inline, internal, external • CSS selectors: universal, element, class, ID, group selectors • Text and font styling: font-family, size, weight, style • Backgrounds, gradients, borders, margin, padding • Positioning: static, relative, absolute, fixed, sticky • Creating navigation bars, dropdown menus and simple layouts **UNIT-IV** **Introduction to JavaScript** • Features, tokens, variables, data types, operators, and expressions • Control constructs, strings arrays, functions • Core language objects, client-side objects • Event handling, Applications related to client-side form validation • Built-In Objects: • The String Object • The Math Object • The Date Object • User Defined Objects
11	**Suggested Resources for Reading** 1. Wanger & Wyke. JavaScript Unleased, Pearson, Latest reprint NewDelhi. 2. Bayross, Ivan, HTML, DHTML,JAVASCRIPT by BPB,Latest reprint 3. Joseph Lowery, Adobe Dreamweaver CS6 Bible byWiley India **Note**: Latest edition of text books may be used.

<table>
<tr><th colspan="2">Course Title</th><th>Advance Routing and Switching</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-505</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>120 hours of practical
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>-</td><td>-</td><td>80</td><td>20</td></tr>
</table>
The practical exam of this course will be of 3 hours duration.
i. There will be no objective type questions.
ii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iii. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
Instructions for practical paper setters and candidates:
i. Examiner will set total of four questions covering the whole syllabus.
ii. Student will attempt any three questions.
iii. Each question carries 10 marks.
iv. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Administrator</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>Routing and Switching technology provide an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Explain IPv4/IPv6 addressing concepts, including private/public addressing, VLSM, summarization, and IPv6 address types to meet LAN/WAN network requirements.
CO2. Configure and verify essential network services such as DHCP, ACLs, NAT, NTP, Syslog, and SNMP to support efficient operations, monitoring, and security.
CO3. Configure and verify network device security features including passwords, SSH, VTY configurations, and secure management access to ensure device protection.
CO4. Configure and verify WAN technologies such as PPP, Frame Relay, and PPPoE, along with external authentication mechanisms, to support reliable wide-area connectivity.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>10</td><td colspan="2">Course Content
UNIT-I
IP Addressing (IPv4/IPv6)
• Explain the purpose of private and public IPv4 addresses.
• Describe how IPv4 addressing operates in LAN/WAN environments.
• Identify scenarios requiring private vs. public IP usage.</td></tr>
</table>

	• Understand the structure and types of IPv6 addresses. • Select appropriate IPv6 addressing schemes for LAN/WAN requirements. **UNIT-II** **Advanced IPv4 & IPv6 Addressing** • Apply VLSM to design efficient IPv4 addressing plans. • Implement route summarization to optimize routing in networks. • Describe dual-stack operation for simultaneous IPv4 and IPv6 usage. • Explain various IPv6 address types (global unicast, multicast, link-local, unique local). • Understand IPv6 auto-configuration methods including EUI-64 **UNIT-III** **Network Device Security** • Configure and verify device password protection mechanisms. • Differentiate between enable and enable secret passwords. • Secure remote access using SSH and disable insecure services like Telnet. • Configure VTY lines and implement management access control. • Apply physical and service password security measures on network devices. **UNIT-IV** **WAN Technologies & Authentication** • Describe external authentication methods for secure network access. • Configure and verify basic WAN serial connections. • Implement and test PPP connections between Cisco routers. • Configure and troubleshoot Frame Relay on Cisco routers. • Set up and troubleshoot PPPoE for broadband connectivity.
11	**Suggested Resources for Reading** 1. Lacoste, CCNP Routing and Switching TSHOOT 300-135: Official Cert Guide, Pearson India. 2. Todd Lammle, CCNA Routing and Switching Study Guide, Wiley India Pvt. Ltd. 3. CCNA Cisco Certified Network Associate Study Guide (3rd Edition), McGraw Hill Education. 4. Mcmillan, Cisco Networking Essentials, John Wiley & Sons. 5. Cryptography and Network Security (3rd Edition), McGraw Hill Education. 6. Cisco: A Beginner's Guide (5th Edition), McGraw-Hill/Osborne Media. Note: Latest edition of text books may be used.

Semester-VI

<table>
<tr><th colspan="2">Course Title</th><th>Internet of Things (IoT) in Networking</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-601</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 4 marks each.
iv. The students are required to attempt one long question of 16 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Administrator</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course introduces students to IoT concepts, architectures, smart devices, communication technologies, protocols, and data analytics, enabling them to understand, design, and apply IoT solutions in real-world applications.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1: Understand IoT fundamentals, architectures, and the role of IT-OT convergence.
CO2: Identify and compare IoT devices, sensors, and connectivity technologies.
CO3: Apply IoT network, transport, and application-layer protocols for communication.
CO4: Explain IoT data analytics, machine learning basics, and major real-world IoT applications.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.</td></tr>
</table>

10	**Course Content** **UNIT - I** Introduction to IoT & IoT Architecture: Definition and evolution of IoT, IoT in digital transformation, convergence of IT & OT, IoT impact: connected roadways, factories, buildings, smart creatures, challenges, and benefits. IoT architectures: oneM2M, World Forum architecture, 7-Layer IoT reference model: Physical devices, Connectivity, Edge computing, Data accumulation, Data abstraction, Applications, Collaboration & processes. . **UNIT - II** IoT Smart Objects & Connectivity Technologies: Sensors, actuators, MEMS, Smart objects: definition, features, trends, Wireless sensor networks (WSNs), Communication requirements: range, frequency, power, topology, IoT access technologies: IEEE 802.15.4 (PHY, MAC, security, topology), IEEE 802.15.4e/g variations, IEEE 1901.2a, IEEE 802.11ah (HaLow), LoRaWAN (MAC, security, topology), NB-IoT, LTE-M, LTE Cat 0, Comparison of short-range, long-range, and cellular IoT connectivity **UNIT - III** Network and Transport Layer Protocols: Constrained nodes and constrained networks, 6LoWPAN (IPv6 Low-power Wireless Personal Area Networks), RPL (Routing Protocol for Low-Power and Lossy Networks). Application Layer Protocols: MQTT (Message Queuing Telemetry Transport), CoAP (Constrained Application Protocol), SCADA (Supervisory Control and Data Acquisition), HTTP/HTTPS and WebSockets for IoT. **UNIT - IV** IoT Data & Analytics: Structured vs unstructured data, Data in motion vs data at rest, IoT data analytics lifecycle and challenges, Machine learning basics: supervised, unsupervised, neural networks, Predictive analytics, BIG data technologies: NoSQL, Hadoop, Kafka, Edge analytics, fog analytics, distributed analytics, Network analytics and Flexible NetFlow. IoT Applications: IoT in manufacturing, IoT in utilities (smart meters, smart grids), IoT in smart and connected cities, IoT in transportation (fleet, traffic), IoT in mining, IoT in public safety.
11	**Suggested Resources for Reading** 1. IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things by David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton, and Jerome Henry (Cisco Press) 2. Internet of Things - Architecture and Design Principles by Raj Kamal 3. Internet of Things: A Hands-On Approach by Bahga & Madisetti 4. Fundamentals of IoT by Sudip Misra & Anandarup Mukherjee Note: Latest edition of text books may be used.

<table>
<tr><th colspan="2">Course Title</th><th>Cyber Security and Data Protection</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-602</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 4 marks each.
iv. The students are required to attempt one long question of 16 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Administrator</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>To introduce students to cyber security concepts, cyber threats, vulnerabilities, cyber laws, and data protection, enabling them to understand, identify, and respond to cybercrimes while ensuring safe and secure digital practices.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1: Understand the fundamentals of cyberspace, cyber security concepts, key terminologies, and the challenges in securing digital environments.
CO2: Identify and analyze different types of web attacks and network vulnerabilities.
CO3: Explain various cybercrimes, their classifications, legal frameworks, and the role of national organizations in cybercrime investigation and cyber security.
CO4: Apply principles of data privacy and data protection</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.</td></tr>
</table>

10	**Course Content** **UNIT - I** Introduction to Cyber security: Defining Cyberspace and Overview of Computer and Web-technology, Architecture of cyberspace, Communication and web technology, Internet, World wide web, Advent of internet, Internet infrastructure for data transfer and governance, Internet society, Regulation of cyberspace, Concept of cyber security, Issues, and challenges of cyber security. Defining Cyber Security, Cyber security terminologies- Cyberspace, attack, attack vector, attack surface, threat, risk, vulnerability, exploit, exploitation, hacker, Controls, Authentication, Access Control and Cryptography. **UNIT-II** Web attack: Browser Attacks, Web Attacks Targeting Users, Obtaining User or Website Data, Email Attacks. Network Vulnerabilities: Overview of vulnerability scanning, Open, Port / Service Identification, Banner /Version Check, Traffic Probe, Vulnerability Probe, Vulnerability Examples, Open VAS, Meta sploit. Networks Vulnerability Scanning (Netcat, Socat), Network Sniffers, and Injection tools. **UNIT-III** Cyber-crime and Cyber law: Classification of cyber-crimes, Common cyber-crimes- cyber-crime targeting computers and mobiles, cyber-crime against women and children, financial frauds, social engineering attacks, malware and ransom-ware attacks, zero day and zero click attacks, Cybercriminals modus-operandi, Reporting of cybercrimes, Remedial and mitigation measures, Legal perspective of cybercrime, IT Act 2000 and its amendments, Cybercrime and offences, Organizations dealing with Cybercrime and Cyber security in India. **UNIT-IV** Data Privacy and Data Security: Defining data, meta-data, big data, non-personal data. Data protection, Data privacy and data security, Personal Data Protection Bill and its compliance, Data protection principles, Big data security issues and challenges, Data protection regulations of other countries- General Data Protection Regulations (GDPR), 2016 Personal Information Protection and Electronic Documents Act (PIPEDA), social media- data privacy and security issues. Firewalls and Packet Filters, password Cracking, Key loggers and Spyware, Virus and Warms, Trojan and backdoors, Steganography, DOS and DDOS attack, SQL injection, Buffer Overflow, Attack on wireless Networks.
11	**Suggested Resources for Reading** 1. Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives by Sumit Belapure and Nina Godbole, Wiley India Pvt. Ltd 2. Security in the Digital Age: Social Media Security Threats and Vulnerabilities by Henry A. Oliver, Create Space Independent Publishing Platform. 3. Cybercrime and Digital Forensics: An Introduction" by Marjie T. Britz (3rd Edition, 4. Security and Privacy in Communication Networks" by Nosseir, Ekici, and Cavallaro 5. Auditing IT Infrastructures for Compliance by Martin Weiss, Michael G. Solomon, 2nd Edition, Jones Bartlett Learning. **Note**: Latest edition of text books may be used.

<table>
<tr><th colspan="2">Course Title</th><th>Firewall Technology</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-603</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practical
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4X 2 marks) = 8 marks
(4X8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Administrator</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>To provide students with practical knowledge of firewall concepts, types, configuration, and administration. The course enables learners to manage network security, web protection, NAT, intrusion prevention, and monitoring through firewall tools.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Explain the need for firewalls, their types, filtering techniques, and how network/application layer firewalls operate.
CO2: Configure and manage firewall settings using Web Admin tools such as dashboard, system settings, access control, certificates, and backup/restore.
CO3: Set up network definitions and apply security features like intrusion prevention, anti-DoS, and country blocking.
CO4: Implement web protection features and analyze logs for security monitoring.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
</table>

10	Course Content **UNIT - I** Firewall: Introduction, why do we need Firewall, How Firewall works, Network Layer Firewall, Application Layer Firewall, Windows Firewall, Types of firewalls: Hardware and Software Firewalls, Firewall Filtering Techniques: Packet Filter, Application Gateway, Circuit-level Gateway, Proxy Server. Installation of Firewall: key functions during installation, special options during installation, installing UTM, Basic Configuration, Backup Restoration. **UNIT-II** Web Admin: Web Admin Menu Button bar, lists searching in lists, dialog boxes, buttons and icons, object lists. Dashboard: dashboard settings, flow monitor, Management: System settings, organizational, hostname, time and date, scan settings, reset configuration or passwords. Web admin settings: general, access control, user rights HTTPs certificate, user preferences, Licensing, up2date, backup/restore, user portal, notifications, hardware and software requirements, status system status configuration, shutdown and restart. **UNIT-III** Definitions & Users: Network definitions, MAC address definitions, service definitions, time period definitions, users and groups, global settings Interfaces and routing, Network Services: DNS: global, forwarders, request routing, static entries, DynDNS, DHCP: Servers, Relay, DHCPv6 Relay, Static Mappings, IPv4 table, Network Protection: Firewall, Rules, Country Blocking, Country Blocking Exceptions, ICMP, Advanced, NAT: Masquerading, NAT, Advanced Threat Protection: Global, Intrusion Prevention: Global attack patterns, Anti-Dos/Flooding, Anti-Portscan, Exceptions, Advanced, Server Load Balancing: Balancing Rules **UNIT-IV** Web Protection: Web filtering: some key differences, common tasks, migration, Global, Https, Policies: Filter action wizard, categories, websites, downloads, antivirus, Web filter Profile: Filter profiles, filter actions, parent proxies, Filtering options: exceptions, websites, bypass users, potentially unwanted applications categories, Logging & Reporting : View Log Files, Today's Log Files, Archived Log Files, Search Log Files, Hardware Daily, Weekly, Monthly, Yearly.
11	**Suggested Resources for Reading** 1. Hucaby, "CCNP Security FIREWALL 642-618 Official Cert Guide", Pearson India 2. Burns, "CCNP Security IPS 642-627 Official Cert Guide", Pearson India 3. Wilkins, "CCNP Security Secure 642-637 Official Cert Guide", Publisher: Pearson India 4. Dameon D Welch-Abernathy, "Essential Check Point Firewall-1 NG: An Installation, Configuration, and Troubleshooting Guide", Pearson Education **Note:** Latest edition of text books may be used.
	BHN-603P (Practical Based on BHN-603) Note: Practical Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation. 1. Examiner will set total of four questions covering the whole syllabus. 2. Student will attempt any three questions. 3. Each question carries 5 marks. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

<table>
<tr><th colspan="2">Course Title</th><th>Artificial Intelligence and Data Science</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-604</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practical
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
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<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4X 2 marks) = 8 marks
(4X8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Administrator</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>To introduce students to the basic concepts of artificial intelligence and data science and enable them to apply fundamental AI and data analysis techniques to solve simple real-world problems.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Understand the basic concepts and scope of artificial intelligence and data science.
CO2. Apply simple AI techniques such as search, classification, and prediction.
CO3. Collect, clean, and analyze data using basic data science methods.
CO4. Use essential tools and libraries for AI and data analysis.
Apply AI and data science concepts to solve basic real-world problems.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>10</td><td colspan="2">Course Content
UNIT-I
Introduction to Artificial Intelligence (AI): Definition, Types of AI, History of AI, Applications of AI in</td></tr>
</table>

	healthcare, education and agriculture, Introduction to key technologies of AI: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Robotics; Ethical issues and Future of AI. Problem Solving: AI problems, Agents and Environments, Structure of Agents. **UNIT-II** Data Science: Introduction to Data Science, evolution and need for Data Science, Types of Data, Data Science Applications in Various Fields, Data Science Life Cycle or Project Stages, Data collection, Data Pre-Processing Overview - Data Cleaning, Data Integration and Transformation, Data Reduction - Data Discretization, Data Storage and management, careers in Data Science **UNIT-III** Managing Data with Spreadsheet Software: Creating a database, sorting& filtering data, custom sorting and advanced filtering, defining, and grouping database ranges, applying data tools like data validation, removing duplicates, text to columns, Flash fill; Managing data using Math, statistical, Trigonometric, String/Text, Date and Time, logical and Database functions, Nested functions. **UNIT-IV** Data Visualization, Searching and Summarizing using Spreadsheet Software: Understanding Chart types, Creating Embedded Charts, formatting Charts: Changing Chart types, adding Titles, Legends and Gridlines, Printing Charts, searching data using HLOOKUP and VLOOKUP, summarizing data using pivot table, analyzing data using What-if Analysis: goal seek, scenario manager and data Table.
11	Suggested Resources for Reading 1. D.Venu Gopal, Data Science and Artificial Intelligence for Beginners Paperback 2. Sudhir Reddy Navar, Introduction to AI/ML and Data Science Paperback 3. C. Raju, Data Science: A Beginner's Guide Paperback 4. Sharma, A, Essentials of AI and Soft Computing, PHI Learning, 2024 5. Kevin Knight, Elaine Rich, B. Nair - Artificial Intelligence, TMH 6. Manish Nigam, Data Analysis with Excel Paperback, BPB Publications 7. Mali, L., Libre Office 5.1 Writer, Calc: Math Formula Book- Vol 1. Notion Press. 8. L. Winston Wayne, Microsoft Excel: Data Analysis & Business Model Paperback **Note**: Latest edition of text books may be used.
	BHN-604P (Practical Based on BHN-604) Note: Practical Paper will be set at the time of examination. Due weightage may be given to the Practical note-book and Assignments in evaluation. 1. Examiner will set total of four questions covering the whole syllabus. 2. Student will attempt any three questions. 3. Each question carries 5 marks. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

<table>
<tr><th colspan="2">Course Title</th><th>Core Python Concepts</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-605</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>120 hours of practical
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>-</td><td>-</td><td>80</td><td>20</td></tr>
</table>
The practical exam of this course will be of 3 hours duration.
i. There will be no objective type questions.
ii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iii. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
Instructions for practical paper setters and candidates:
v. Examiner will set total of four questions covering the whole syllabus.
vi. Student will attempt any three questions.
vii. Each question carries 10 marks.
viii. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Administrator</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>The objective of the module is to make the student understand the basic concepts of python programming language, lists, tuples, classes, and dictionary. The module will enable students to do programming using the python library.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. The fundamental concepts of python programming, syntax, and semantics.
CO2. Using Lists, tuples, Dictionaries, functions in python.
CO3. Creating and using Files in python.
CO4. Using python modules and functions.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>10</td><td colspan="2">Course Content
UNIT - I:
Introduction to Python Programming
• Data Types
• Operators & Expressions
• Variables, Scope of a variable (Local, Global)
• Type Conversion (Implicit & Explicit)
Control Flow Statements
• Simple if, if-else, if-elif-else chain, if statement with lists
Loops
• Selection, Indentation, Repetition
• Break and Continue statements</td></tr>
</table>

	• Nested Loops • while loop, with lists, with dictionaries **Strings** • String Operations • Traversal of Strings • String Methods • Inbuilt String Functions **UNIT - II** **Lists, Tuples, Dictionaries** • List Operations, Traversal of Lists, List Methods • Inbuilt List Functions • Nested Lists, Copying Lists • Lists as Arguments in Functions • Mutable vs Immutable Data Types **Tuples** • Tuple Operations, Traversal of Tuples, Tuple Methods • Inbuilt Tuple Functions, Nested Tuples **Dictionaries** • Dictionary Traversal, Dictionary Methods • Inbuilt Dictionary Functions • Manipulating Dictionaries (Add, Update, Delete) **UNIT - III** **Functions and Classes** **Functions** • User-defined Functions, Defining, Passing arguments • Return values, Passing Lists to Functions • Python Standard Libraries **Classes (Object-Oriented Programming)** • Creating and Using a Class • Working with Classes, Instances • Importing Classes • Python Standard Library for OOP **UNIT - IV** **Exception & File Handling** **Exception Handling** • Exceptions and Errors in Python • try-except block • else block • Handling Specific Exceptions: Zero Division Error, File Not Found Error **File Handling** • Types of Files • Opening and Closing Files • Writing to a File • Reading from a File • Setting File Offset (seek()) • Creating a File • Traversing a File
11	**Suggested Resources for Reading** 1. Learning Python, Mark Lutz 5th Edition, O'Reilly Media, Inc., 2015 2. Python: Python Programming for Beginners - The Comprehensive Guide To Python Programming: Computer Programming, Computer Language, Computer Science, Adam Stark, CreateSpace

	Independent Publishing Platform, 2016 3. Python Essential Reference, David M Beazley, 4th edition, Pearson Education Inc., 2009. 4. Python Programming: An Introduction to Computer Science, John M Zelle, 2nd Edition ,2003 5. Python Crash Course, A Hands-On, Project-Based Introduction to Programming, Eric Maththes, 2nd Edition, No Starch Press, 2019 **Note**: Latest edition of text books may be used.

<table>
<tr><th colspan="2">Course Title</th><th>Advance IP Routing</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-606</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>120 hours of practical
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
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<tr><td>-</td><td>-</td><td>80</td><td>20</td></tr>
</table>
The practical exam of this course will be of 3 hours duration.
i. There will be no objective type questions.
ii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iii. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
Instructions for practical paper setters and candidates:
i. Examiner will set total of four questions covering the whole syllabus.
ii. Student will attempt any three questions.
iii. Each question carries 10 marks.
iv. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Administrator</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>IP ROUTING technology provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Implement dynamic routing solutions using EIGRP and OSPF by analysing network requirements.
CO2. Design and deploy eBGP and IPv6-based routing solutions, including IPv4-IPv6 interoperability.
CO3.
Configure and validate redistribution across multiple routing domains by execution plans, and verification documents.
CO4. Implement Layer 3 path control and remote connectivity solutions by configuring path manipulation, broadband technologies, and branch-access technologies.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>10</td><td colspan="2">UNIT - I
EIGRP & OSPF
• Fundamentals and network requirements of EIGRP.
• EIGRP implementation planning steps.
• Configuring EIGRP routing on routers.
• Verifying EIGRP using show/debug commands.</td></tr>
</table>

	• Fundamentals and network requirements of multi-area OSPF. • Configuring and verifying OSPF routing. • Documentation of EIGRP and OSPF results. **UNIT - II** **eBGP & IPv6** • Basics and requirements for implementing eBGP. • eBGP implementation and verification planning. • Configuring eBGP routing and validating with show/debug. • Basics and requirements of IPv6 deployment. • IPv6 routing and IPv4-IPv6 coexistence configuration. • Verifying IPv6 implementation with appropriate commands. • Documentation of eBGP and IPv6 configurations. **UNIT - III** **Route Redistribution** • Purpose and need for IPv4/IPv6 route redistribution. • Planning redistribution based on network analysis. • Configuration of route redistribution in routers. • Managing metrics, filtering, and loop prevention. • Verifying redistribution using show/debug commands. • Documentation of redistribution plan and results. • Differences between IPv4 and IPv6 redistribution. **UNIT-IV** **Path Control & Branch Services** • Need and planning for Layer 3 path control. • Configuring path control techniques (PBR, metrics). • Verifying proper implementation of path control. • Basics of teleworker and branch connectivity services. • Understanding broadband access technologies. • VPN technologies and GRE tunnel configuration. • Branch access technologies and WAN connectivity.
11	**Suggested Resources for Reading** 1. CCNP: Routing, CCIE No Kevin Wallace, Sean Odom, Todd Lammle 2. CCNP Routing and Switching V2.0 Official Cert Guide Library, Author Kevin Wallace, David Hucaby, Raymond Lacoste, Wendell Odom, Publisher CISCO System, 2014 3. CCNP: ROUNTING and SWITCHING QUICK Reference, Author: Denise Donohue, Brent Stewart, Publisher Cisco press 4. CCNP: ROUTE 642-902, Author: Wendell Odom, Publisher: Cisco press **Note:** Latest edition of text books may be used**.**